Algebraic Formulation of the Collatz Operator in Jacobsthal Space: The $3J + J(0,1)$ Transformation

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Abstract

In previous studies, we demonstrated that the Collatz orbit can be reproduced by calculating the sum of adjacent terms in the Jacobsthal (J) sequence. This paper advances that approach by algebraically formulating the scalar operation of "multiply by 3 and add 1" ($3n + 1$) in the Collatz process as an affine transformation of a spinor (column vector) in J-space. Specifically, defining the spinor of an odd node b in J-space as $J(a, b)$, we show that the Collatz operation can be naturally represented by the following vector operation:

$$\boldsymbol{3J(a,b) + J(0,1)}$$

This formulation clarifies the algebraic behavior of the additive noise "+1," which is the source of difficulty in the Collatz conjecture, and reveals an algebraic mechanism associated with the trivial loop. This study provides a fundamental algebraic framework for expanding and analyzing the Collatz space as a dynamical system.

1. Introduction

In previous studies, we demonstrated that the Collatz orbit can be reproduced by calculating the sum of adjacent terms in the Jacobsthal (J) sequence $J(a, b)$:

$$b_{next} = (a + b)/2^k$$

Here, for any odd number b, we calculate its internal state a using the formula below, and denote the expansion into a J-sequence as $J(a, b)$:

$$a = (b + 1)/2$$

$$J(a, b) = a, b, 2a + b, 2a + 3b, 4a + 5b, \ldots$$

However, to discuss the algebraic structure of the Collatz orbit in a unified manner, a universal operator is required. Therefore, we define the basic Collatz operation $n \mapsto 3n + 1$ as a fixed affine transformation applied to a spinor vector in J-space as follows:

$$3J(a,b) + J(0,1)$$

Furthermore, to represent the "division by 2" operation of the Collatz process within J-space, we introduce the concept of "selection of the reduced spinor."

2. Vector Definition of J-Spinor

We define any odd number b as a two-component column vector (J-Spinor) in J-space, consisting of an internal state a and an observed node b:

$$J(a,b) = \begin{pmatrix} a \\ b \end{pmatrix}$$

In the initial state, the two components satisfy the following fundamental relation:

$$b = 2a - 1$$

While the standard Collatz operation applies $3b + 1$ solely to b, we extend this operation to the entire J-Spinor.

3. The $3J + J(0,1)$ Transformation

We define the odd step of the Collatz operation as the following affine transformation applied to the J-Spinor:

$$3\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3a \\ 3b + 1 \end{pmatrix}$$

The lower component (the b-component) becomes $3b + 1$, perfectly matching the actual Collatz orbit. Conversely, the upper component (the a-component) only receives the addition of 0 from the additive vector $(0,1)^T$, ensuring that the additive noise $+1$ is excluded, leaving a pure $3a$. This operator functions as a filter that isolates the additive noise exclusively within the b-component.

4. Algebraic Neutralization and Phase Shift

When division by 2^k (the shift count) is performed after this vector operation, the vector for the next state is defined as follows:

$$\begin{pmatrix} a_{next} \\ b_{next} \end{pmatrix} = \frac{1}{2^k}\begin{pmatrix} 3a \\ 3b+1 \end{pmatrix} = \begin{pmatrix} \frac{3a}{2^k} \\ \frac{3b+1}{2^k} \end{pmatrix}$$

Expanding the expression for b_{next} by substituting the fundamental relation $b = 2a - 1$ yields:

$$b_{next} = \frac{3(2a-1)+1}{2^k} = \frac{6a-2}{2^k} = 2\left(\frac{3a}{2^k}\right) - \frac{2}{2^k}$$

Here, the first term on the right side contains $a_{next} = \frac{3a}{2^k}$, determined by the vector operation. Thus, the following relational equation is obtained:

$$b_{next} = 2(a_{next}) - \frac{1}{2^{k-1}}$$

When $k = 1$, the offset is $1/2^0 = 1$, and the integer structure $b_{next} = 2(a_{next}) - 1$ is maintained. However, when $k \geq 2$, a phase shift occurs in the offset, indicating that fractional values are introduced into a_{next}.

5. J-space 2-adic Reduction and Orbit Analysis

How, then, can we avoid the fractions caused by shifts of $k \geq 2$ and maintain the integer structure $b = 2a - 1$ in J-space? In a Collatz orbit, groups (branches) of odd nodes converging to the same subsequent odd node form an equivalence class. Here, we define the set of J-sequences sharing the same reduced spinor as a **J equivalence class**. For example, all odd values appearing at even indices in the following $J(0,1)$ sequence reach 1 through the Collatz operation and are considered to belong to the same J equivalence class:

$$J(0,1) = 0, 1, 1, 3, 5, 11, 21, 43, 85, \ldots$$

$$J(0,1)_{even} = 1, 5, 21, 85, \ldots$$

Therefore, to avoid the fractions generated by shifts of $k \geq 2$, instead of continuing the calculation directly, one must reselect the reduced spinor located at the root of this branch as the representative element. In this paper, we call this process **J-space 2-adic Reduction**. For instance, the reduced spinor of the $J(0,1)$ sequence is $J(0,1)$.

Using this method, we verified the entire orbit from the initial node $b_0 = 7$ to the terminal node 1 (**Table 1**).

Table 1. Collatz Orbit for Initial Value 7 in J-Spinor Representation

Step	Odd Node	Expanded J	Reduced J (Input)	3J+J(0,1)	k	Mapped J	Reduced J (Output)
1	7	J(4, 7)	J(4, 7)	J(12, 22)	1	J(6, 11)	J(6, 11)
2	11	J(6, 11)	J(6, 11)	J(18, 34)	1	J(9, 17)	J(4, 9)
3	17	J(9, 17)	J(4, 9)	J(12, 28)	2	J(3, 7)	J(2, 3)
4	13	J(7, 13)	J(2, 3)	J(6, 10)	1	J(3, 5)	J(0, 1)
5	5	J(3, 5)	J(0, 1)	J(0, 4)	2	J(0, 1)	J(0, 1)
6	1	J(0, 1)	J(0, 1)	J(0, 4)	2	J(0, 1)	J(0, 1)

The J-spinor transition process of the Collatz orbit from the initial node $b_0 = 7$ to the fixed point $J(0,1)$ (node 1). Each column represents the following algebraic operations:

- **Odd Node:** The current odd node in the standard Collatz orbit.
- **Expanded J:** The J sequence generated from the current odd node and its internal state $a = (b + 1)/2$ before J-space reduction.
- **Reduced J (Input):** The reduced spinor extracted as the representative element of the corresponding J equivalence class, serving as the input vector for the current step.
- $\boldsymbol{3J + J(0,1)}$**:** The result of applying the affine transformation to the input spinor.
- $\boldsymbol{k}$**:** The number of shifts applied (equivalent to division by 2^k in the standard Collatz operation).
- **Mapped J:** The normalized odd spinor obtained through the Shift & Cut normalization process without introducing fractional components.
- **Reduced J (Output):** The reduced spinor extracted from Mapped J, serving as the input for the next step.

Step 1: $7 \to 11$ $(k = 1)$

The initial state is $J(4,7)$.

$$3\begin{pmatrix}4\\7\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}12\\22\end{pmatrix} \xrightarrow{\div 2^1} \begin{pmatrix}6\\11\end{pmatrix}$$

Since $k = 1$, we obtain $11 = 2(6) - 1$, maintaining the integer structure.

Step 2: $11 \to 17$ $(k = 1)$

We apply the same calculation to $J(6,11)$.

$$3\begin{pmatrix}6\\11\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}18\\34\end{pmatrix} \xrightarrow{\div 2^1} \begin{pmatrix}9\\17\end{pmatrix}$$

Again, since $k = 1$, the structure is maintained.

Step 3: $\mathbf{17 \to 13}$ $(\boldsymbol{k = 2})$

Applying the transformation directly to $J(9,17)$ results in a fraction ($a = 27/4$) due to the $k = 2$ shift.

$$3\begin{pmatrix}9\\17\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}27\\52\end{pmatrix} \xrightarrow{\div 2^2} \begin{pmatrix}27/4\\13\end{pmatrix}$$

To avoid the fraction, we expand our view from a single node to the underlying J-sequence. Expanding 17 into its J-sequence yields $J(4,9) = \{4,\ 9,\ 17, \ldots\}$, and its reduced spinor is $J(4,9)$. Therefore, we apply the transformation to this representative element.

$$3\begin{pmatrix}4\\9\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}12\\28\end{pmatrix} \xrightarrow{\div 2^2} \begin{pmatrix}3\\7\end{pmatrix}$$

The calculation yields $J(3,7)$, which expands to $J(2,3) = \{2, 3, 7, 13, \ldots\}$. The expanded J-sequence of 13, the next odd node after 17, is also $J(2,3) = \{2, 3, 7, 13, \ldots\}$. This calculation suggests that the operation not only landed without passing through fractions, but also algebraically projected (mapped) the entire expanded J-sequence of 17 onto the expanded J-sequence of 13.

Step 4: $\mathbf{13 \to 5}$ $(\boldsymbol{k = 3})$

Applying the transformation directly to $J(7,13)$ generates a fraction ($a = 21/8$) due to the $k = 3$ shift. Thus, we select the reduced spinor $J(2,3)$ from the expanded J-sequence of 13: $J(2,3) = \{2, 3, 7, 13, \ldots\}$.

$$3\begin{pmatrix}2\\3\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}6\\10\end{pmatrix} \xrightarrow{\div 2^1} \begin{pmatrix}3\\5\end{pmatrix}$$

The shift count is minimized to $k = 1$, landing directly on the integer spinor $J(3,5)$ for $b = 5$.

Step 5: $\mathbf{5 \to 1}$ $(\boldsymbol{k = 4})$

Applying the transformation directly to $J(3,5)$ generates a fraction ($a = 9/16$) due to the $k = 4$ shift. Thus, we select the reduced spinor $J(0,1)$ from the expanded J-sequence of 5: $J(0,1) = \{0, 1, 1, 3, 5, \ldots\}$.

$$3\begin{pmatrix}0\\1\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\4\end{pmatrix}$$

The resulting vector can be factored by the scalar 4.

$$\binom{0}{4} = 4\binom{0}{1} = 4J(0,1)$$

Since $b = 4$ is divisible by 2^2, applying a shift down (multiplication by $1/4$) returns it to the original $J(0,1)$.

$$\frac{1}{4} \cdot 4\binom{0}{1} = \binom{0}{1} = J(0,1)$$

The trivial loop "$1 \rightarrow 4 \rightarrow 1$" associated with the terminal state originates from the fact that $J(0,1)$ is the eigenvector (fixed point) of this transformation, with an eigenvalue of 4 (i.e., 2^2).

6. Conclusion

In this paper, we formulated the Collatz operation as the $3J + J(0,1)$ transformation in J-Spinor space. This transformation possesses the following characteristics:

1) **Phase Shift:** Continuing the calculation directly results in fractions when $k \geq 2$.
2) **Selection of the Reduced Spinor:** Fractions are avoided and the integer structure is maintained by selecting the reduced spinor from the odd node's expanded J-sequence (J-space 2-adic Reduction).
3) **Fixed-Point Structure:** The terminal point $J(0,1)$ is the eigenvector of this transformation, appearing as the algebraic representation of the trivial loop.

This study provides a fundamental algebraic framework for expanding and analyzing the Collatz space as a dynamical system.

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